

WASP-77Ab

R-1317

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-77_211217 · created 2026-07-31T19:11:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459565.64909 +/- 0.00054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.069 +/- 0.022
Transit depth (Rp/Rs)^2	0.47 +/- 0.3 [%]
Orbital Inclination (inc)	88.56 +/- 1.42
Airmass coefficient 1 (a1)	0.9963 +/- 0.0094
Airmass coefficient 2 (a2)	0.001 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	#3 - [222, 168]
Optimal Aperture	4.23
Optimal Annulus	17.46
Transit Duration (day)	0.0845 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.069 ± 0.022 vs 0.1301 (deviation 1.88σ)
Duration fit vs published	0.0845 d vs 0.0908 d (deviation 1.71σ)
Mid-time O-C	-0.1 min
Depth SNR	2.1
Residual scatter	0.94 %

Light curve

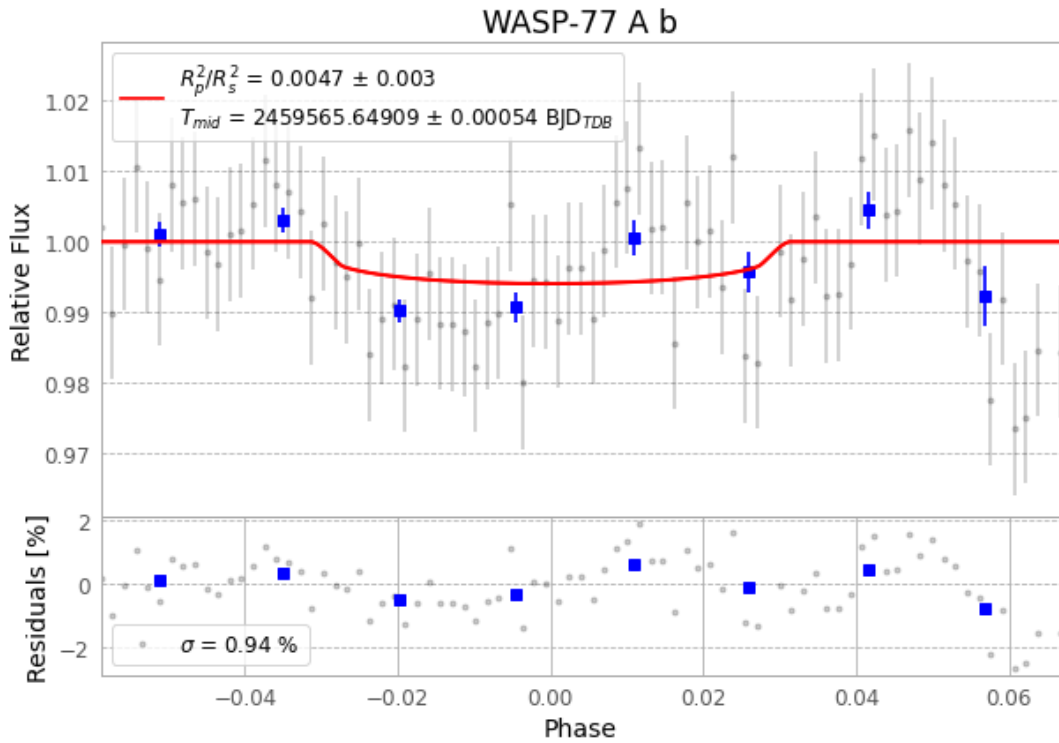


Figure 1 — FinalLightCurve_WASP-77 A b_2021-12-16.png (SHA-256 2f6daa265016cad1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [539, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a963a16ec9d7d6d3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-77211217013314.FITS → WASP-77211217053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-211217.log (f7f69944bb7e...), FinalLightCurve_WASP-77 A b_2021-12-16.png (2f6daa265016...), FinalLightCurve_WASP-77 A b_2021-12-16.csv (b90fafafcab8...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:11Z.